

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 10%

QUESTIONS:2

Total Marks:20

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I D.Manjunadha_(192324237)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.Manjunadha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

Q1)

Problem:

- Respond to real-time events accurately.
- Learn and adapt to user preferences.
- Maintain user comfort by controlling lighting and temperature.
- Improve energy efficiency by minimizing unnecessary power consumption.
- Ensure home security by detecting unusual activities and triggering alerts.
- Make optimal decisions when multiple objectives, such as comfort, energy savings, and security, must be balanced.

Hybrid Intelligent Agent

A Hybrid Intelligent Agent combining Model-Based, Goal-Based, and Utility-Based approaches is the most effective solution.

Working of the Hybrid Agent

1. Model-Based Component
 - Maintains room occupancy, temperature history, and device status.
2. Goal-Based Component
 - Ensures target temperature, energy savings, and home security.
3. Utility-Based Component
 - Selects the best action by balancing comfort, energy consumption, electricity cost, and user preferences.

1. Simple Reflex Agent:

decisions only based on the current input using predefined IF–THEN rules

Advantages:

- Fast response.
- Easy to design and implement.
- Low computational cost.

Limitations:

- Cannot remember previous events.
- Cannot learn user preferences.
- Inefficient in complex environments.

2. Model-Based Agent:

Agent maintains an internal model of the environment. It stores previous states and uses memory to make better decisions.

Advantages:

- Uses memory for decision-making.
- Works effectively in partially observable environments.
- More reliable than a simple reflex agent.

Limitations:

- Requires additional memory.
- More complex than a simple reflex agent.

3. Goal-Based Agent:

A Goal-Based Agent selects actions that help achieve predefined goals.

Advantages:

- Supports planning and intelligent decision-making.
- Flexible in different situations.
- Achieves long-term objectives.

Limitations:

- Requires planning algorithms.
- Higher computational complexity.

4. Utility-Based Agent:

Agent evaluates multiple possible actions and chooses the one that provides the highest overall benefit based on a utility function.

Advantages:

- Balances multiple objectives.
- Learns user preferences over time.
- Optimizes system performance.

Limitations:

- Most complex agent type.
- Requires utility calculations and learning algorithms.

Comparison of Intelligent Agents

agent type	memory	planning	learning	suitable for smart home
simple reflex agent	no	no	no	basic automation tasks
model-based agent	yes	no	limited	occupancy and environment monitoring
goal-based agent	yes	yes	limited	achieving comfort and security goals
utility-based agent	yes	yes	yes	optimizing comfort, energy efficiency, and security

Result:

For a smart home automation system, a Hybrid Intelligent Agent is the most suitable solution. While Simple Reflex Agents provide quick responses, they cannot learn or adapt. Model-Based Agents remember past states, Goal-Based Agents plan actions to achieve objectives, and Utility-Based Agents optimize decisions by balancing comfort, energy efficiency, and security. Combining these approaches results in an intelligent, adaptive, and efficient smart home system.

2Q)

Problem Statement:

A robot is placed in an unknown maze and must find the exit without prior knowledge of the maze layout. It needs to explore the environment, avoid obstacles, and identify a valid path using uninformed search algorithms. The challenge is to achieve efficient navigation with minimum cost, time, and memory usage.

1. Uniform Cost Search (UCS):

Working

- ucs expands the node with the lowest cumulative path cost.
- it uses a priority queue to store nodes according to their total path cost.
- if movement costs differ, ucs always finds the least-cost path.
- the search continues until the exit is reached.

Example

If moving through some cells costs more energy than others:

S --1--> A --2--> G

```
\      \  
4      1  
\      \  
B --1--> G
```

Possible Paths:

- $S \rightarrow A \rightarrow G = \text{Cost } 3 \text{ least}$
- $S \rightarrow B \rightarrow G = \text{Cost } 5$

UCS selects $S \rightarrow A \rightarrow G$ because it has the minimum cost.

2. Iterative Deepening Search (IDS):

Working

- ids repeatedly performs depth-limited search (dls).
- it begins with depth limit 0, then 1, then 2, and so on until the goal is found.
- it combines the advantages of depth-first search and breadth-first search.

Example

Depth = 0

S

Depth = 1

S

/ \

A B

Depth = 2

S

/ \

A B
 /\
 C G
 Goal G is found at depth 2.

3. Advantages And Disadvantages

feature	uniform cost search (ucs)	iterative deepening search (ids)
complete	yes	yes
optimal	yes (for positive costs)	yes (equal step costs)
time	$O(b^{(1+c*/\epsilon)})$	$O(b^d)$
space	$O(b^{(1+c*/\epsilon)})$	$O(bd)$
memory usage	high	low

Intelligent Agents In The Maze

different intelligent agents behave differently while exploring the maze.

a. simple reflex agent

- makes decisions only using the current perception.
- no memory of previous locations.
- may repeatedly visit the same path.
- not suitable for unknown mazes.

example

wall ahead → turn right

b. model-based reflex agent

- maintains an internal map of explored areas.
- remembers visited locations.
- avoids revisiting the same paths.
- better than a simple reflex agent.

c. goal-based agent

- has a clear goal: reach the exit.
- uses search algorithms to evaluate possible paths.
- chooses actions that move closer to the goal.
- well suited for maze navigation.

Best:

goal-based agent and uniform cost search (ucs)

- a goal-based agent focuses on reaching the exit using planned actions rather than reacting blindly.
- ucs guarantees the least-cost path when movement costs vary (such as energy usage or difficult terrain).
- the agent continuously updates its knowledge while searching and selects the most cost-effective route.

- this combination improves navigation efficiency, minimizes travel cost, and ensures an optimal solution.

if every movement has the same cost and memory is limited, a goal-based agent + iterative deepening search (ids) is also an excellent choice because ids uses much less memory while still finding the shortest path in terms of steps.